

Chapter 3

Functional Anatomy of the Lumbar Spine

You have most likely already studied the basic anatomy of the spine. This chapter begins by revisiting some anatomical features, possibly in a way not previously considered. Then these features are related to normal function and injury mechanics to lay the foundation for the prevention and rehabilitation strategies that follow. I believe that clinicians and scientists alike who specialize in low back disorders do not devote sufficient effort to simply considering the anatomy. The answers to many questions relevant to the clinician can be found within an anatomical framework, wherein lies the mechanical foundation for preventing and rehabilitating back disorders. The understanding of injury mechanisms presented here will help ensure that you do not unknowingly include injury-exacerbating maneuvers in therapeutic exercise prescriptions.

Anatomy Trains

Many students of anatomy study cadavers and learn the names of the parts, but never learn their functional relevance. The cadaver shows the parts in a static and obviously dead state. This trite statement is a segue into the notion that a living body moves dynamically and that muscles influence joints other than the ones they cross. Thomas Myers wrote the groundbreaking book *Anatomy Trains* (2014), which introduced the notion of linkages between muscles and connective tissues that span many joints. Muscles, fascia, and other connective tissues transmit force along their length in a series arrangement. This is conventional thought. However, forces are also transmitted laterally, in a parallel sense to adjacent structures. On one hand, these anatomy trains create great efficiency in economy of movement and strength augmentation, but on the other, pathology in one component translates along the linkage affecting distant normal tissues.

Another logical extension of this concept of anatomical linkages is the greater functional role of many muscles and ligaments. For example, consider the gluteal muscles. The best anatomy student will state that they extend and externally rotate the femur via the hip joint. This is correct, yet naive. When a person is standing and squatting, using the gluteal muscles to extend the hips also extends the knees; thus, the gluteal muscles become powerful knee extensors. Poor use of the gluteal muscles eventually leads to knee overload and pain. We can now proceed with a discussion as anatomists, biomechanists, clinicians, and coaches.

Basic Neural Structure

The greatest athletes who are elite performers, and who avoid injury, are very wise in understanding the process of activating muscles and groups of muscles. They are masters at using imagery to control the motor unit recruitment process. We study these athletes to learn their processes so that we can employ these techniques with people who have painful backs. The basics of what you need to know about neural integration, some of which are introduced here, can be found in wonderful resources such as Kandel, Schwartz, and Jessell (2000).

Motion may occur from a conscious thought in the brain that instigates muscle activation, or from a more subconscious process involving an encoded pattern thought to reside in the spinal cord. Traumatic events can recode these patterns to perturbed states, as can chronic and acute pain. Rerecoding these perturbed patterns back to normal is an issue addressed in part III of this book.

Better links between training and neuroanatomy, neurophysiology, and rehabilitation can be found in my textbook *Ultimate Back Fitness and Performance* (McGill, 2014). Several relevant discussions illustrate, for example, why machines cannot create the many variations of force development within a muscle to stimulate all motor units. In the torso, for example, the oblique muscles have many neuromuscular compartments that must be stimulated with demand. Slow and isolationist approaches typical of bodybuilding do not offer a rich proprioceptive environment that provides variable motion, balance, force projection, and direction challenges involving the full linkage.

Another aspect for designing the best manual therapies is based in neuroanatomy. Although David Butler's book *The Sensitive Nervous System* (2000) is a wonderful neuroanatomical source, an introduction to the concepts is attempted here. Often, pain that is attributed to muscle turns out to be neurogenic. The tests described later in this book are founded on several principles. Anatomically, the spinal cord and all nervous tissues linked in series (lumbar nerve roots, the sciatic nerve, and so on) can be tensioned, released, mobilized, and flossed with specific and coordinated joint motions. Tensioning nerves only causes more pain—neurogenic pain cannot be stretched away. Sadly, too many patients with so-called tight hamstrings or

sciatic symptoms pursue stretching programs that produce only temporary relief. This relief results from the activation of the stretch reflex in the back extensor muscles, but it typically lasts only about 20 minutes. The pain and stiffness return. It is often possible to break the cycle by replacing the stretching with neural mobilization. Butler described well the benefits of mobilizing a nerve along its entire tract together with the mechanics that create local tensions. With coordinated cervical, hip, knee, and ankle motion, the lumbar nerve roots, cauda equina, and sciatic tract can be mobilized and flossed without tensioning (shown in chapter 10). We have too many cases of intransigent sciatica cured with this approach to ignore it.

Vascular Anatomy

All spinal tissues have a vascular supply with the exception of the inner disc. The implications of the avascular nucleus are described in the section Intervertebral Disc later in this chapter. The curious case of vertebral veins is introduced here. The veins leaving the vertebral bodies are the only veins in the body known to lack valves. Venous valves prevent the backflow of blood. But as we will see shortly in the discussion of vertebral mechanics, this anatomic feature is critical. It appears that the veins act as a hydraulic outlet from the vertebral body, enabling the expulsion of blood under high compressive loading. In this way both the arteries and the veins may provide a protective mechanism and the ultimate hydraulic shock-dampening system.

Vertebrae

As you undoubtedly know, the spine has 12 thoracic and 5 lumbar vertebrae. The construction of each vertebral body may be likened to a barrel with round walls made of relatively stiff cortical bone (see [figure 3.1](#)). The top and bottom of the barrel are made of a more deformable cartilage plate (end plate) that is approximately 0.6 mm (0.02 in.) thick but thinnest in the central region (Roberts, Menage, and Urban, 1989). The end plate is porous to allow for the transport of nutrients such as oxygen and glucose, whereas the inside of the barrel is filled with cancellous bone. The trabecular arrangement within the cancellous bone is aligned with the stress trajectories that develop during activity. Three orientations dominate—one vertical and two oblique (Gallois and Japoit, 1925) (see [figure 3.2](#)).

Figure 3.1 The parts of a typical lumbar vertebra. (a) Lateral view, (b) superior view, and (c) posterior view.

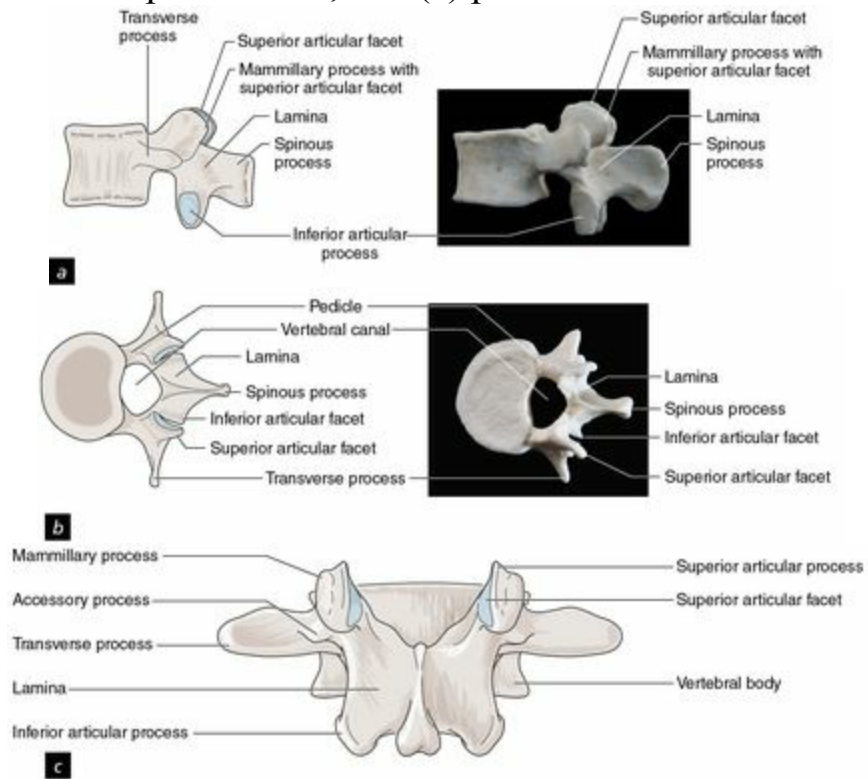
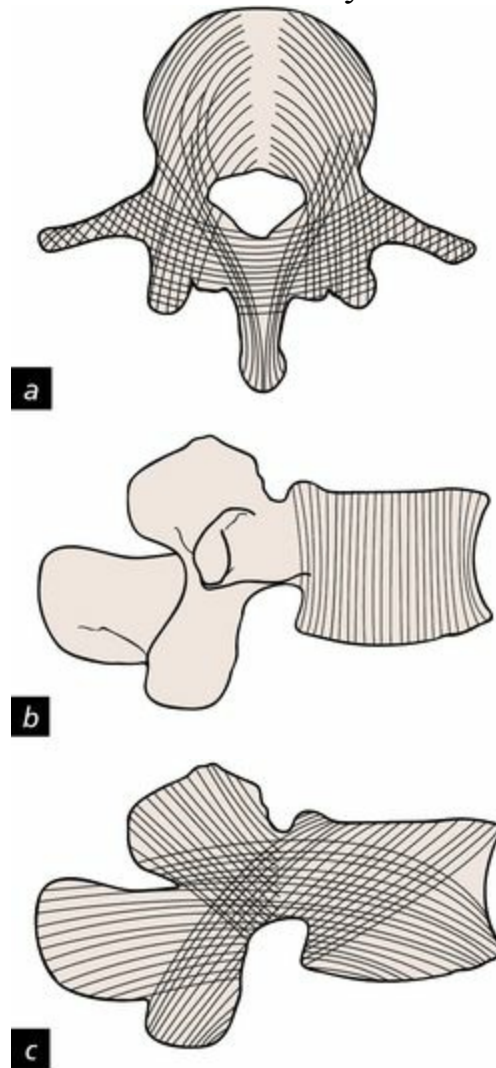


Figure 3.2 The arrangement of the trabeculae (first noted by Gallois and Japoit in 1925) is aligned with the dominant trajectories of stress. (a , b, c)
The three trabecular systems.



Vertebral Architecture and Load Bearing

The very special architecture of the vertebral bodies determines how they bear compressive load and fail under excessive loading. The walls of the vertebrae (or sides of the barrel) remain rigid upon compression, but the nucleus of the disc pressurizes (the classic work is by Nachemson, 1960, 1966) and causes the cartilaginous end plates of the vertebrae to bulge

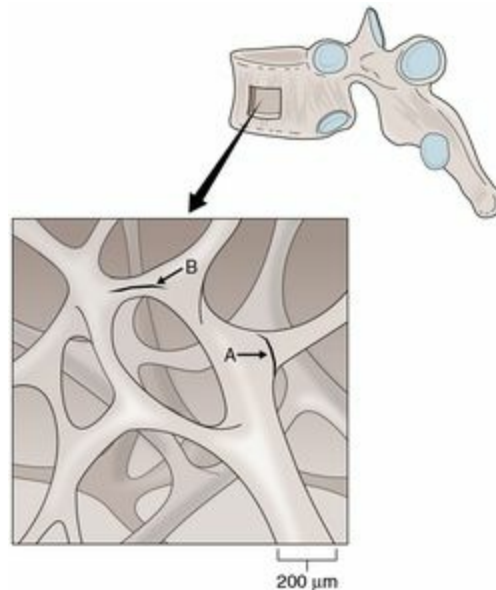
inward, seemingly to compress the cancellous bone (Brinckmann, Biggemann, and Hilweg, 1989). In fact, under compression the cancellous bone fails first (Gunning, Callaghan, and McGill, 2001), making it the determinant of failure tolerance of the spine (at least when the spine is not positioned at the end range of motion). It is difficult to injure the disc annulus this way (annular failure is discussed later).

Although this notion is contrary to the concept that the vertebral bodies are rigid, the functional interpretation of this anatomy suggests a very clever shock-absorbing and load-bearing system. Farfan (1973) proposed the notion that the vertebral bodies act as shock absorbers of the spine, although he based this more on vertebral body fluid flow than on end-plate bulging. He suggested that the discs were not the major shock absorbers of the spine, contrary to virtually any textbook on the subject. Because the nucleus is an incompressible fluid, bulging end plates suggest fluid expulsion from the vertebral bodies, specifically blood through the perivertebral sinuses (Roaf, 1960). This mechanism suggests a protective dissipation upon quasistatic and dynamic compressive loading of the spine. The case study literature abounds with compression fractures of the vertebral body during dynamic loading in which the disc remained intact (e.g., during tobogganing and sledding; Kelly and Robinson, 2003). More vertebral body-based shock-absorbing mechanisms are documented later. In summary, the common statement found in many textbooks that the discs are the shock absorbers of the spine now appears questionable; rather, the vertebral bodies appear to play a dominant role in performing this function.

The notion of deformable vertebrae is a new one for many. How do the end plates bulge inward into seemingly rigid bone? The answer appears to be in the architecture of the cancellous bone. Vertebral cancellous bone structure is dominated by a system of columns of bone (shown in [figure 3.2](#)) that run vertically from end plate to end plate. The vertical columns are tied together with smaller transverse trabeculae. Upon axial compression, as the end plates bulge into the vertebral bodies, these columns experience compression and appear to bend. Under excessive compressive load, the bending columns buckle as the smaller bony transverse trabeculae fracture, as documented by Fyhrie and Schaffler (1994) (see [figure 3.3](#)). In this way, the cancellous bone can rebound back to its original shape (at least 95% of the original unloaded shape) when the load is removed, even after suffering fracture and delamination of the transverse trabeculae. This architecture appears to afford

excellent elastic deformation, even after marked damage, and then to regain its original structure and function as it heals. Damaged cancellous fractures appear to heal quickly, given the small amount of osteogenic activity needed, at least when compared with the length of time needed to repair collagenous tissues.

Figure 3.3 Under compressive loading, bulging of the end plate causes buckling stresses in the vertical trabeculae, which, when excessive, cause damage in the transverse trabeculae. Note (A) the vertical (from compression) and (B) horizontal (from tension) cracks in the transverse trabeculae.



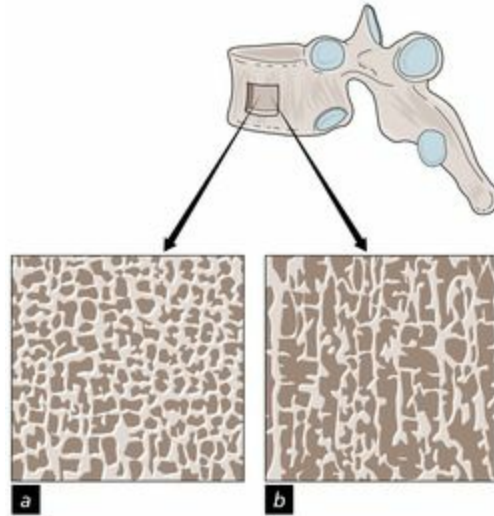
Understanding Vertebral Mechanics

To truly appreciate vertebral behavior, obtain a vertebra from a butcher (bovine or porcine is ideal). Hold the vertebra end plate to end plate between your thumb and finger and squeeze. If you have never done this before, you will be amazed by the deformation and elasticity. The vertebra experiences similar deformation as the incompressible nucleus of the disc presses over the central end plate during spine compression *in vivo*.

Microfracturing of the trabeculae can occur with repetitive loading at levels well below the failure level from a single cycle of load. Lu and colleagues (2001) demonstrated that cyclic loading at 10% of ultimate failure load caused no damage or change in stiffness; but with 20,000 cycles of load at 20 to 30% of the ultimate failure load, both stiffness and energy absorbed at failure were decreased. Highly repetitive loads, even at quite low magnitudes, appear to cause microdamage.

The osteoporotic vertebra is characterized by mineral loss and declining bone density in the trabeculae. Because transverse trabeculae are far fewer in number than longitudinal trabeculae, and because they are generally of smaller diameter, the transverse trabeculae specifically are the target for mechanical compromise with osteoporotic mineral loss (Silva and Gibson, 1997) (see [figure 3.4](#)). Interestingly, the same authors noted a higher tendency for the transverse trabeculae to disappear in females with greater incidence than in males. This loss in mechanical integrity of the transverse trabeculae has a great influence on the compressive strength of the vertebrae via the mechanism described earlier. Thus, the osteoporotic vertebra begins to slowly collapse when exposed to excessive load, with serial buckling of the columns of bone, ultimately developing the classic wedge shape.

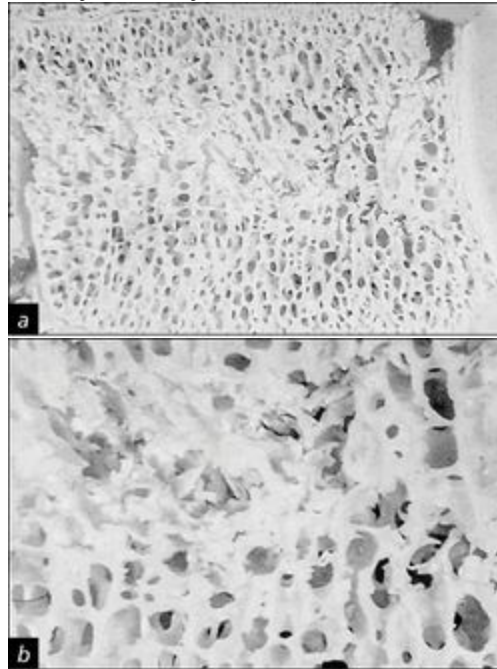
Figure 3.4 With aging, the transverse or horizontal trabeculae thin and eventually lose their ability to support the vertical trabeculae, which can then buckle, causing vertebral collapse. (a) A healthy trabecular bone network from a 47-year-old woman. (b) Perforation in a horizontal trabecula in an elderly woman.



It is interesting to contrast the other extreme of the bone density spectrum. The transverse trabeculae harvested from specimens who performed heavy work (in particular, weightlifters) were thick and dense. In addition, where the transverse trabeculae intersected with the vertical columns, the joints were characterized by heavy bony gusseting, similar to what a welder would weld to strengthen a right-angled joint. The transverse trabeculae appear to be crucial in determining compressive strength.

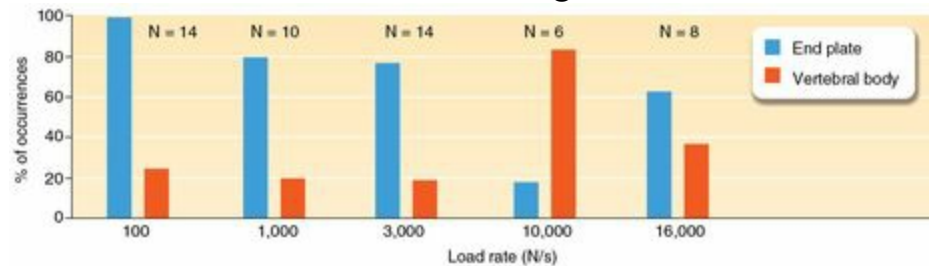
In yet another observed failure mechanism of the vertebral body, termed the slow crush, extensive trabecular damage is observed without concomitant loss of stiffness or abrupt change in the load–deformation relationship (Gunning, Callaghan, and McGill, 2001) (see [figure 3.5](#)). Because slope change in the load–deformation relationship is often used to identify the yield point, or the initial tissue damage, this injury can go unnoticed. Interestingly, vertebrae failing under ever-increasing compressive load gradually increase in stiffness, a testament to the wonderful architecture of the transverse and vertical trabeculae. Also interesting is the load-rate dependence of bone strength: the end plate appears to fail first at low load rates, whereas the vertebral bony elements fail at higher load rates (see [figure 3.6](#)).

Figure 3.5 (a) Massive trabecular damage found during the dissection and removal of marrow following an excessive slow crush compressive load. (b) Higher magnification of the crush fracture. Even this massive fracture was not clear on X-ray or any other examination method.



Reprinted from Clinical Biomechanics, Vol. 16(6) J. Gunning, J. Callaghan, and S. McGill, “The role of prior loading history and spinal posture on the compressive tolerance and type of failure in the spine using a porcine trauma model,” 471-480. Copyright 2001, with permission of Elsevier.

Figure 3.6 Compression injuries at various load rates. At low rates of compressive load, the end plate appears to be the first structure to fail, but bone will fracture first under higher rates of load.



Both the disc and the vertebrae deform while supporting spinal loads. Under excessive compressive loading, the bulging of the end plates into the vertebral bodies also causes radial stresses in the end plate sufficient to cause fracture in a stellate pattern (see [figure 3.7](#)). These fractures, or cracks, in the end plate are sometimes sufficiently large to allow the liquid nucleus to squirt through the end plate into the vertebral body (McGill, 1997) (see [figures 3.8](#) and [3.9](#)). Sometimes a local area of bone collapses under the end plate to create a pit or crater that goes on to form the classic Schmorl's node (see [figure 3.10](#)). This type of injury is associated with compression of the spine when the spine is within the neutral range of motion (i.e., not flexed, bent, or twisted). In my experience, this type of compressive injury is very common and often misdiagnosed as a herniated disc because of the flattened interdiscal space seen on planar X-rays. However, note that in end-plate fractures, the annulus of the disc remains intact. It is simply a case of the nucleus leaving the disc and progressing through the end plate into the cancellous core of the vertebra (sometimes referred to as a vertical herniation). Over the years we have compressed over 400 spinal units in a neutral posture, and all but 2 resulted in end-plate fractures as the primary tissue damage.

Figure 3.7 A stellate-patterned end-plate fracture (indicated with arrow) occurs as the nucleus is pressurized under compressive load, which causes it to bulge the end plate, imparting tensile stresses.

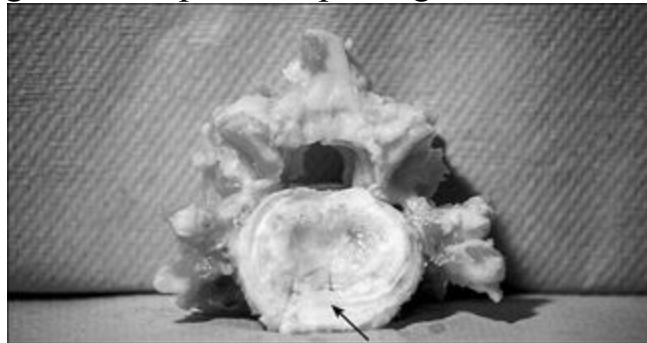


Photo from Stuart McGill

Figure 3.8 Under compressive loading, the nucleus pressurizes, causing the end plate to bulge into the vertebral body. With excessive radial-tensile stress, the end plate will fracture and the viscous nucleus will squirt through the crack into the vertebral body.

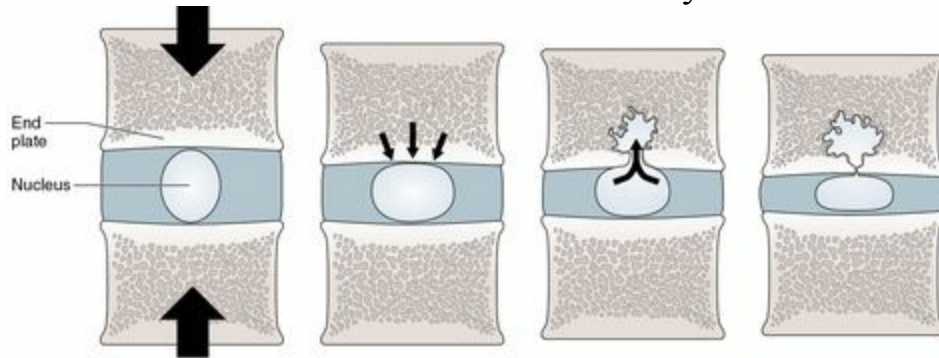


Figure 3.9 Following a more severe end-plate fracture, a portion of the nucleus has squirted through into the vertebral body (shown at the tip of the scalpel and the arrow). This is a porcine specimen.



Photo from Stuart McGill

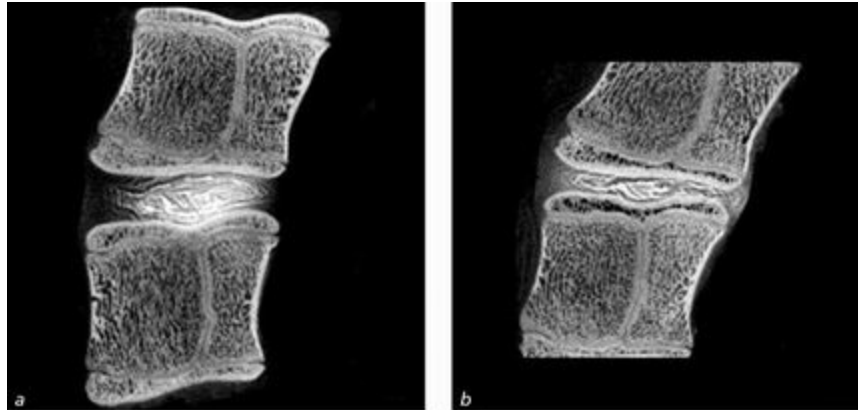
Figure 3.10 In addition to end-plate fractures, other injuries to the end plate under compressive load include pits, which occur as trabecular bone fractures in small areas under the end plate. These go on to form Schmorl's nodes.



Photo from Stuart McGill.

Our most recent work has mimicked compressive loads similar to those imposed during heavy lifting, such as performing deadlift exercises. Using micro-CT imaging, we documented a crushing of the trabecular framework across the thin growth plate (see [figure 3.11](#)) (Balkovec et al., 2014). Although this damage would be clinically important, the resulting weakening of the end plate would make it susceptible to secondary end-plate fracturing and Schmorl's nodes.

Figure 3.11 Micro-CT images reveal the crushed growth plate from compressive loading mimicking a very heavy lift. (a) A virgin spine specimen prior to mechanical loading. (b) The same specimen after undergoing a compressive load with a flexion bending protocol mimicking a heavy deadlift session. Note the distinct regions directly beneath the end plate where the trabecular bone appears to have fractured and the density is reduced.



Reprinted from Journal of Biomechanical Engineering, Vol. 35(6), C. Balkovec, J. Vernengo, and S.M. McGill, “The use of a novel injectable hydrogel nucleus pulposus replacement in restoring the mechanical properties of cyclically fatigued porcine intervertebral discs,” pp.: 61004-61005, copyright 2014, with permission of Elsevier.

Osteophytes are a characteristic of older spines classed as degenerated. Michael Adams' group (Al-Rawahi et al., 2011) assessed whether these are degenerative or adaptive. Removing them reduced the ability of the joint to withstand bending loads by up to one-half. Losses in compression resistance were less than 20%. They concluded that osteophytes are an adaptation to joint instability, which stimulates their formation. Further, the presence of osteophytes increased measures of bone mineral density, but such measures may actually underestimate vertebral compressive strength.

End-Plate Fractures

In my experience, end-plate fracture with the loss of nuclear fluids through the crack into the vertebral body (often forming Schmorl's nodes) is a very common compressive injury and perhaps the most misdiagnosed. Loss of the disc nucleus results in a flattened interdiscal space that, when seen on planar X-rays, is usually diagnosed as a herniated disc or degenerated disc.

However, the annulus of the disc remains intact. It's simply a case of the nucleus squirting through the end-plate crack into the cancellous core of the vertebra. True disc herniation requires very special mechanical conditions that are described shortly. When compressing spines in the lab, we hear an audible pop at the instant of end-plate fracture—exactly what patients report when they describe details of the event that resulted in their pain. I also strongly suspect that some fractures are somewhat benign in that no immediate severe pain occurs. Others, however, may be instantaneously acute; it depends on the biomechanical changes that accompany the fracture. If there is substantial loss of the nucleus from the disc (i.e., it is vertically herniated), then immediate loss of disc height and subsequent compromise of nerve root space will result. At this point the end-plate fracture will mimic the symptoms of true herniation—another reason for the common misdiagnosis.

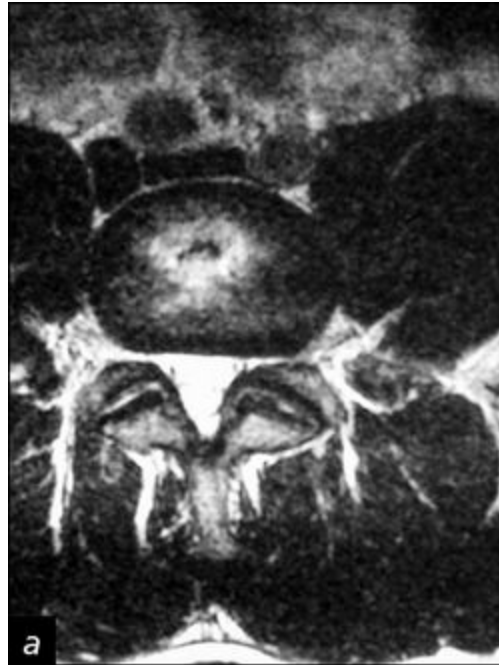
Modic Changes and What They Mean

MRI images sometimes show signals in the vertebral body along the end plate. These signals, called Modic changes after Dr. Modic (Modic et al., 1988), have sparked much discussion regarding their clinical significance, if any. Our experience strongly suggests that these findings represent edema in the bone. The edema can have two causative pathways. First, an end-plate fracture sufficient to allow nucleus material to leak into the vertebral body sets off a substantial inflammatory response. The immune system and vascular environment respond to the invading nucleus as foreign material, setting off the reaction. This mechanism suggests damage to the cartilaginous end plate. This is consistent with the report of Zhi and colleagues (2014), who noted more hyaline end-plate cartilage particulate in harvested herniated material during surgery in those with Modic changes than in those without.

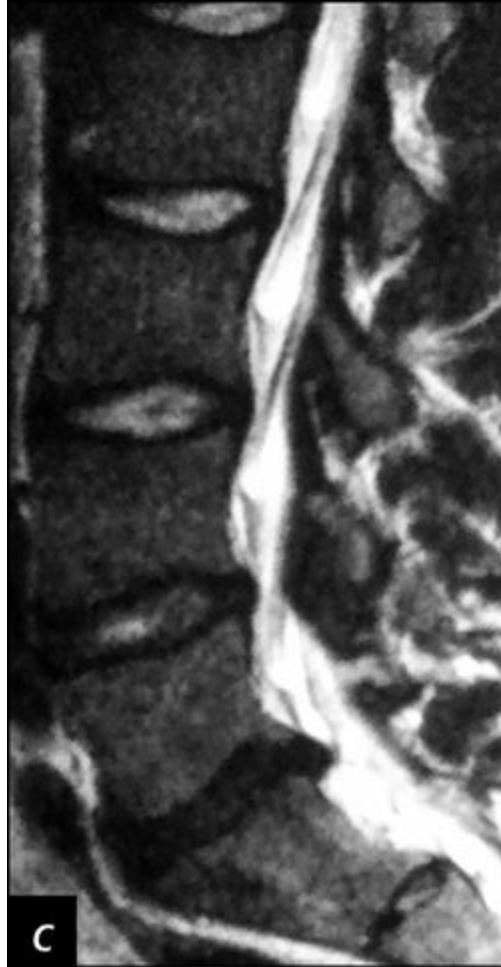
The second causative pathway of edema is consistent with what is called a bone bruise. Several athletes we have seen had well-documented training histories as well as annual records of repeated MRI images. These case studies enabled us to analyze the associations among loading, progressive changes in joint function, and MRI images. A typical example would be a person who shows end-plate damage but no other MRI changes. Over the next few years, the joint slowly flattens and loses water signal in the disc.

Then anterior margins of the vertebrae begin to collide during flexion motion (see [figure 3.12](#)). Modic changes then appear in the bone at these contact points. I refer to these as bone bruises, because pain is now reported central in the spine, at the damaged level upon flexion movement. Once Modic changes have been documented, longer-lasting pain can be expected; our experience suggests that 12 to 18 months is typical. Avoiding this movement reduces pain sensitivity.

Figure 3.12 The history of a professional athlete showing initial end-plate damage in 2009, disc height loss by 2009, and colliding anterior margins of the vertebrae with bone bruise formation by 2011. (a) Schmorl's node at L5, 2008. (b , c, d) Disc bulge, no bone bruise, 2009. (e , f, g) Bone bruise at L5 and S1, retrolisthesis, October 2011.







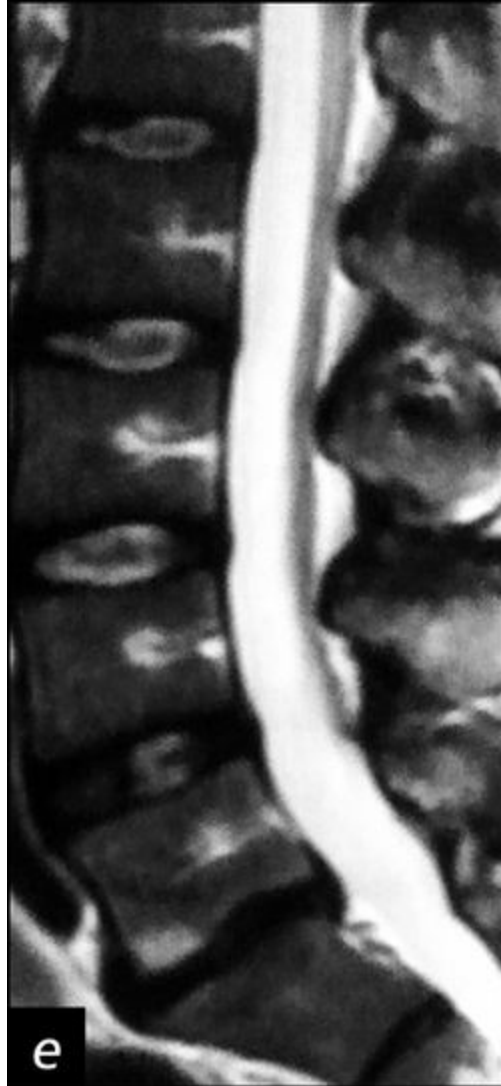






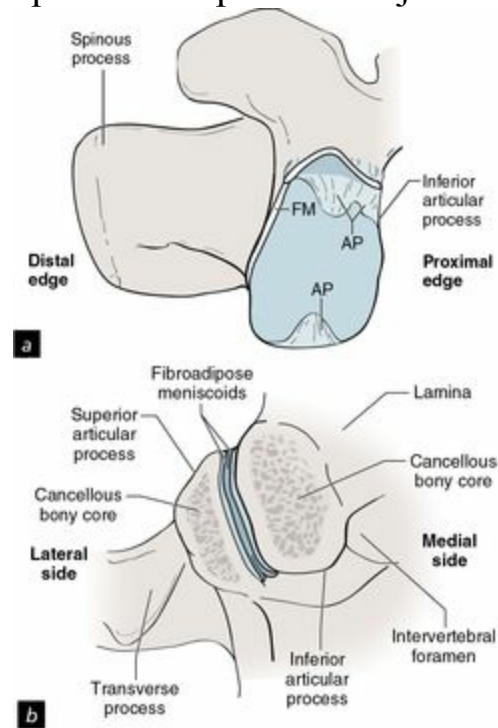
Photo from Stuart McGill

Posterior Elements of the Vertebrae

The posterior elements of the vertebrae (pedicles, laminae, spinous processes, and facet joints) have a shell of cortical bone but contain a cancellous bony core in the thick sections. The transverse processes project laterally together with a superior and an inferior pair of facet joints (see [figure 4.1](#)). On the lateral surface of the bone that forms the superior facets are the accessory and mamillary processes that, together with the transverse process, are major attachment sites of the longissimus and iliocostalis extensor muscle groups (described later). The facet joints are typical synovial joints in that the articulating surfaces are covered with hyaline cartilage and are contained within a capsule. Fibroadipose enlargements, or meniscoids, are found around the rim of the facet, although mostly at the proximal and distal edges (Bogduk and Engel, 1984), which have been implicated as a

possible structure that could bind and lock the facet joint (see [figure 3.13](#), a and b).

Figure 3.13 (a) Lateral view of the inferior articular process, revealing the facet, the fibroadipose meniscoids, and the adipose tissue pad, which have been implicated in joint binding. (b) Cross section of the posterior view of the facet joint, showing the positions of the fibroadipose meniscoids and the adipose tissue pads in the joint.



The neural arch in general (pedicles and laminae) appears to be somewhat flexible. In fact, Bedzinski (1992) demonstrated flexibility of the pars interarticularis during flexion–extension of cadaveric spines. Dickey, Pierrynowski, and Bednar (1996) documented up to 3° changes of the right pedicle with respect to the left pedicle during quite mild daily activities using pedicle screws in vivo. Failure of these elements together with facet damage, leading to spondylolisthesis, is sometimes blamed exclusively on anteroposterior shear forces. However, a case could be made from epidemiological evidence in athletes such as gymnasts and Australian cricket bowlers (Hardcastle, Annear, and Foster, 1992) that the damage to these posterior elements may also be associated with full range of motion. In fact, the faster the bowler, the higher the risk of fatigue fracture (Ranawat, Dowell, and Heywood-Waddington, 2003). The cyclic full spine flexion and extension in these sorts of activities fatigue the arch with repeated stress

reversals.

Flexibility of the Posterior Elements of the Vertebrae

In the preceding sidebar Understanding Vertebral Mechanics, I recommended obtaining a vertebra from the butcher. Grasp the vertebral body in one hand and the whole neural arch in the other. Bend the arch up and down and note the flexibility. This flexural displacement occurs in each cycle of full spine flexion–extension common in events such as women’s gymnastics ([figure 3.15](#)). If this cycling continues, the stress reversals eventually cause a fatigue crack in the pars. Further repetition causes the crack to propagate through the full width of the pars, eventually resulting in fracture—and in vivo, the condition of spondylolisthesis.

Figure 3.15 Repetitive gymnastics moves such as this cause stress–strain reversals in the pars. In sufficient numbers they result in fatigue fracture—leading to spondylolisthesis.



Shariffc | Dreamstime.com - Gymnastics Action Photo

On the other hand, there is no doubt that excessive shear forces cause injury to these posterior vertebral elements. Posterior shear of the superior vertebrae can lead to ligamentous damage but also failure in the vertebra itself, because the end plate often avulses from the rest of the vertebral body (Yingling and McGill, 1999b) (see [figure 3.14](#)). Both our lab observations and discussions with international colleagues have reinforced our suspicion that this type of failure may be more common in the adolescent and geriatric spine than in the young and middle-aged adult spine. Further work is needed for confirmation.

Figure 3.14 Shear injuries include fracture of the facet base and, on occasion, end-plate avulsion from the vertebrae.



Photo from Stuart McGill

Cripton and colleagues (1995) documented that anterior shear of the superior vertebra causes pars and facet fracture leading to spondylolisthesis with a typical tolerance of an adult lumbar spine of approximately 2,000 N. Although similar injury mechanisms and tolerance values were observed in young porcine spine specimens (Yingling and McGill, 1999a, 1999b), the type of injury appeared to be modulated by loading rate. Specifically, anterior shear forces produced soft tissue injury at low load rates (100 N/s), but fractures of the pars, facet face, and vertebral body were observed at higher load rates (7,000 N/s). Posterior shear forces applied at low load rates produced soft tissue failure and vertebral body fracture, whereas those at higher load rates produced wedge fractures and facet damage.

Although shear tolerance of the vertebral motion unit appears to be in the range of 2,000 to 2,800 N for one-time loading, Norman and colleagues (1998) noticed an increase in reported back pain in vivo in jobs that exposed workers to repetitive shear loads greater than 500 N. We consider these the best guidelines currently available.

Neural Arch Fracture

Spondylolisthesis and neural arch defects are endemic among female gymnasts and cricket bowlers, to name a couple of athlete types. Patients with spondylolisthesis generally do not do well with therapeutic exercise that takes the spine through the range of motion; rather, stability should be the

rehabilitation objective. Ranawat, Dowell, and Heywood-Waddington (2003) made a strong case for conservative approaches, which were successful even with cricket bowlers: only a few needed follow-up surgical interventions. I have been involved in litigation cases in which clear spondylolisthesis existed but was alleged to be related to a specific event (e.g., a recent automobile accident). In surgery, however, there was evidence of substantial osteogenic activity, suggesting that the injury was quite old. In fact, these patients (former gymnasts) must have had this damage while competing, but they were so fit and their spines were so stable that they were able to remain in competition. Following retirement and after having children and losing fitness, a rather minor event became the instigator of their symptoms.

A final note is relevant regarding the repeated and prolonged extension postures and motions associated with the McKenzie approach discussed later in this book. Some people have what are called kissing spines, in which adjacent posterior spines collide in full extension at one level. The involved level is usually due to a simple case of anatomical variation. These may become more frequent in people with disc height loss when the posterior spine becomes more approximated. Jim Taylor (Twomey and Taylor, 1987) noted destructive changes in the interspinous ligaments as they are repeatedly crushed at the kissing spine level. He described the changes as the development of a “fibrocartilaginous covering on the colliding bone and a bursa-like cavity surrounded by fat lined with a synovial membrane.” There is no question regarding the efficacy of the McKenzie extension routines for some people with acute discogenic back issues. I am cautious about having people continue to engage in these postures following recovery from the acute episode, however—the spine may pay a price.

Sacroiliac Joints

The sacroiliac (SI) joints are quite large and have little mobility. This means that large forces are often required during provocative testing. There has been much discussion over the years about the existence of measurable movement, whether there should be movement, whether the movement can be asymmetrical between right and left sides, and how much difference is required to be clinically detectable. The majority of research has been focused on test reliability rather than on mechanisms and clinical technique. As noted later, painful symptoms reported in the sacroiliac joint region are often referred from discogenic disorders irritating sciatic nerve roots. After using diagnostic injections, Young, Aprill, and Laslett (2003) stated that SI joint pain rarely occurred above the L5 level, or around the midline; rather, they stated that it was much more likely to occur unilaterally (exacerbated by rising from a chair) and was correlated with three positive SI joint pain provocation tests. However, like all tests, this depends on clinicians' ability to direct force to the joints while sparing surrounding tissues.